ECO WASTE

Team member:

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BUSINESS UNDERSTANDING

01 BUSINESS BACKGROUND

Business Context

Singapore is committed to a sustainable, zero-waste future through efficient waste management and higher recycling rates

Why it matters?

- Waste growing with population and ecocnomic growth
- Achieving Green Plan 2030 and a circular economy
- High landfill use and resource wastage

PROBLEM STATEMENT :

“How can Singapore improve its waste management and recycling efficiency?”

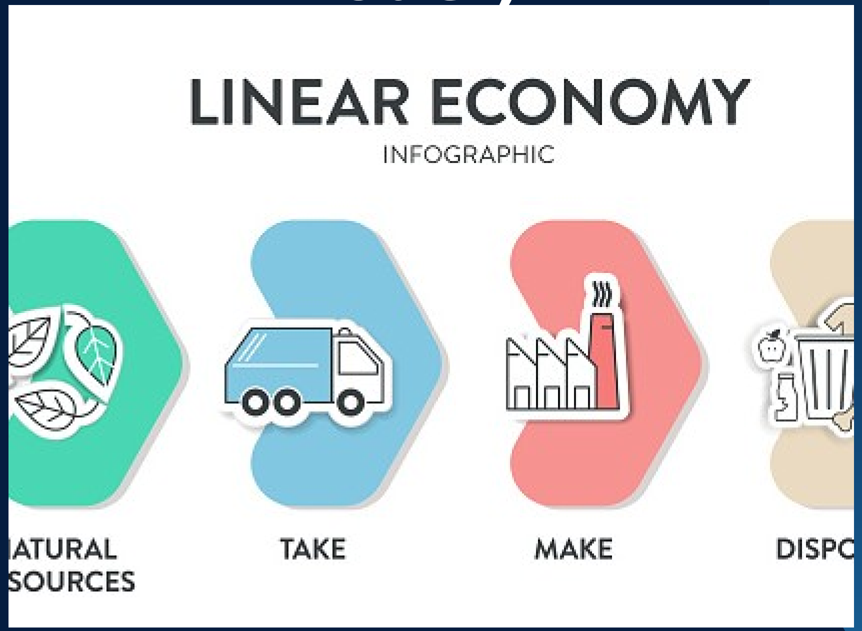
Why is waste and recycling efficiency a problem?

Land Scarcity and the Semakau Landfill Limitation

Low Domestic Recycling Rates

High Incineration Burden and Environmental Impact

Resource Inefficiency (Linear Economy Model)



02

BUSINESS OBJECTIVE

01 EVALUATE

Evaluate Singapore waste and recycling performance

02 Analysis

Identify trends, patterns, and future forecasts,

03 Improve

Recommend solutions based on global best practices.

DATA MINING GOALS

01

Identify major contributors and trends in Singapore's waste generation.

02

Analyse Singapore's recycling rate trends by waste type and identify changes in recycling performance over time

03

Analyse Singapore waste generation rate (per capita) and forecast

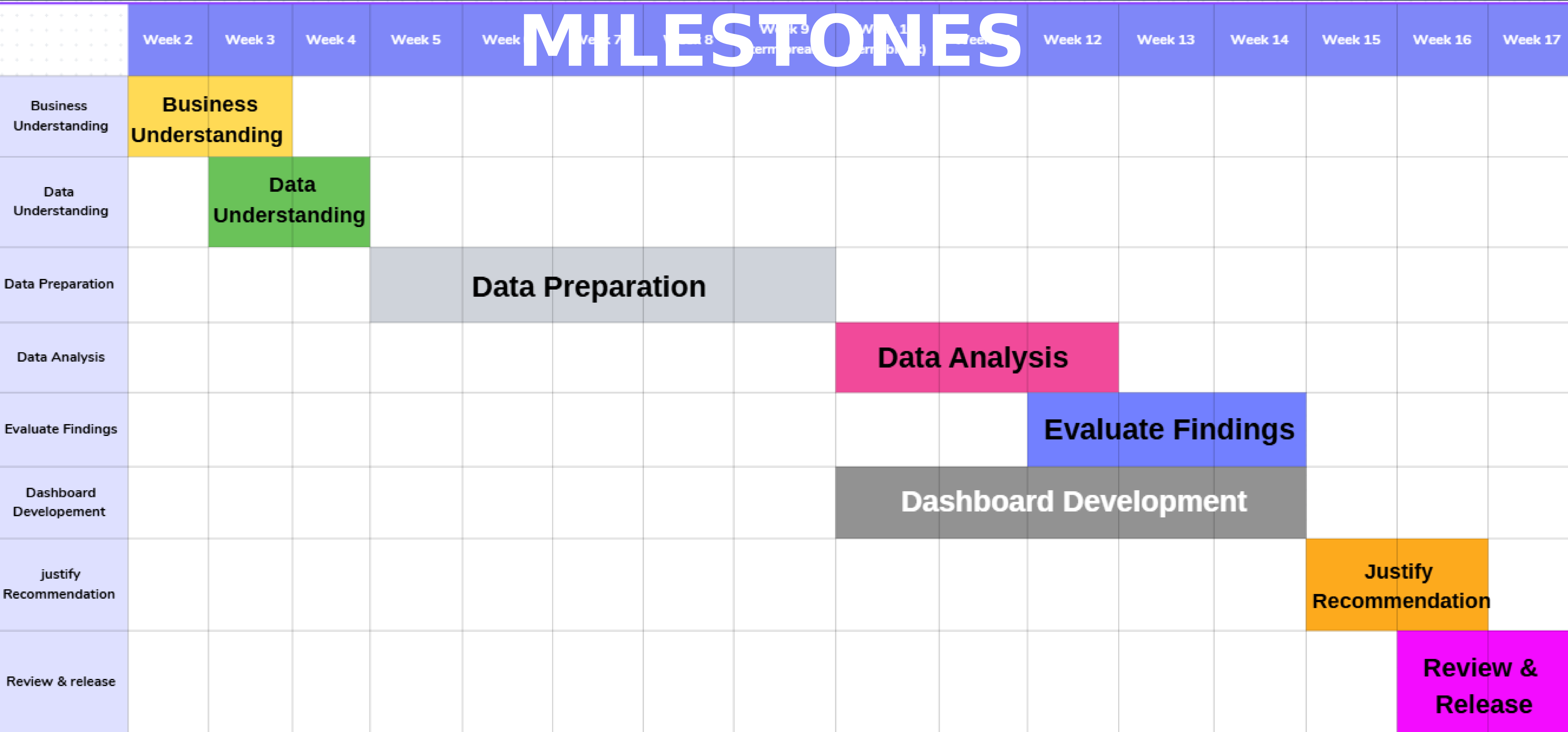
04

Identify and rank OECD countries that achieved the greatest reduction in municipal waste generated per person from 2000–2020, and use the best-performing countries as benchmarks for Singapore’s future waste-reduction scenario.

05

Identify countries that has high overall recycling rate from 2000-2020, and recommend recycling solution to singapore based on the best performing country

PROJECT TIMELINE & MILESTONES



ASSUMPTIONS

- **Data Reliability**

Data obtained from (data.gov) (OECD)
are **accurate and reliable**

- **Forecasting assumption/Singapore can achieve a similar reduction rate to the benchmark countries.**

Current recycling trends are expected to continue in the future

- **Policy assumption**

The selected policy is treated as a key contributor to the country’s waste reduction and recycling rate increase

DATA UNDERSTANDING

07 PUBLIC AVALIABLE DATA

DATA.GOV.SG

- Offical Singapore government open data portal
- Provides local waste management and recycling data
- Up-to-date and relevant to Singapore’s context

Organisation for Economic Co-operation and Development

- International organisation providing standardised data
- Enables comparison of waste and recycling performance across countries
- Supports benchmarking and learning from global best practices

DATA DICTIONARY

Dataset 1: Waste Management and Recycling Statistics of Singapore

Valid format	Description	Data field	Description of values	Business Data Object
Text	Waste Category	DataSeries	Paper/Cardboard, glass....	Singapore waste management and recycling statistic
YYYY	Reporting year	Year	2000-2024	
Numeric (Tonnes)	total waste generated	Total waste generated	amount of waste produced anually	
Numeric (Tonnes)	total waste recycled	Total waste recycled	amount of waste recycles annually	
Numeric (%)	Percentage of total waste recycled	Overall Recycling Rate	Recycling rate of waste	

DATA DICTIONARY

Dataset 2: Singapore annual Population

Business Data Object	Data field	Description	Valid format	Description of values
Indicators on singapore's Population over the years	DataSeries	Population indicator category	Text	Total Population
	Year	Reporting year	YYYY	2000-2025
	Total Population	Total number of people in Singapore	Whole number	6,111,175 (2025)

DATA DICTIONARY

Dataset 3: Oversea Countries Waste Management and Recycling Statistics

Business Data Object	Data field	Description	Valid format	Description of values
Oversea Countries waste management and recycling statistic	REF_AREA	Country	Text	Korea, Germany....
	TIME_PERIOD	Reporting year	YYYY	1995- 2020
	Recycling	Recycling rate	Numeric (%)	Percentage of treated waste
	Waste generated	Total waste generated	Numeric (kg/person)	Amount of waste generated by each person in each year

Emmanuel

DATA QUALITY ISSUES IDENTIFIED

1. Missing data

Some countries had incomplete yearly records, such as Australia, missing data from 2001–2006. With missing values being kept blank .

COUNTRY DATA SUMMARY

	country	rows_in_file	years_with_value	missing_year_count	missing_years	has_all_
0	Canada	0	0	21	2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007...	
1	Costa Rica	11	11	10	2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007...	
2	Australia	15	15	6	2001, 2002, 2003, 2004, 2005, 2006	
3	New Zealand	15	15	6	2000, 2001, 2005, 2007, 2008, 2009	
4	Colombia	16	16	5	2000, 2001, 2002, 2019, 2020	
5	Mexico	17	17	4	2013, 2015, 2017, 2019	
6	Chile	19	19	2	2012, 2013	
7	United States	19	19	2	2019, 2020	

1. Unusual data fluctuations

Some countries' data have big rises or drops causing steep rise and falls in the graphs.

Reference area	TIME_PERIOD	Recycling (%)
Sweden	2018	29.88658
Sweden	2019	32.46299
Sweden	2020	20.31401

DATA CLEANING

Emmanuel

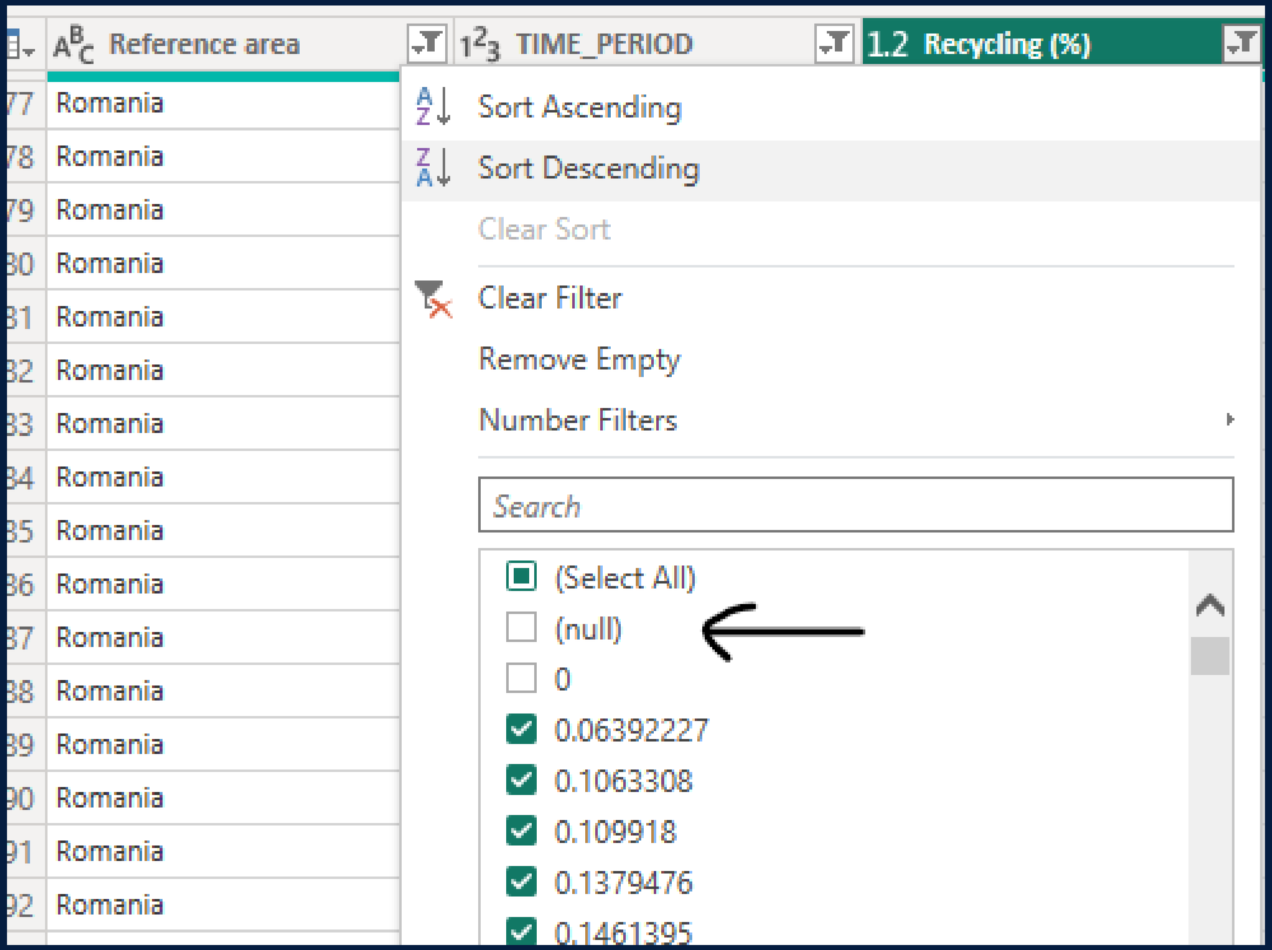
```python
# Use simple column names
rename_columns = {
    "Reference area": "country",
    "TIME_PERIOD": "year",
    "OBS_VALUE": "waste_per_person_kg"
}

if "OBS_STATUS" in oecd_kg.columns:
    rename_columns["OBS_STATUS"] = "kg_status"

if "Observation status" in oecd_kg.columns:
    rename_columns["Observation status"] = "kg_status_desc"

oecd_kg = oecd_kg.rename(columns=rename_columns)

# Keep year as integer
oecd_kg["year"] = oecd_kg["year"].astype(int)

print("Rows after filtering:", len(oecd_kg))
print("Countries found:", oecd_kg["country"].nunique())

display(oecd_kg.head())
```

- **Standardised the data:**

Rename columns and standardise units of measurements (kg) for better analysis.

- **Handle missing and invalid values :**

Identified and exclude missing/null values to ensure data completeness.
(via google colab / powerBi query)



DATA TRANSFORMING

Emmanuel

- **Reshaped the cleaned dataset:**

Converts cleaned data into a format good for trend, comparison, and ranking analysis.

year	country	waste_per_person_2000	waste_per_person_2020	kg_difference_2000_2020	kg_percentage_change_2000_2020
0	Austria	579.8130	833.6866	253.8736	43.785427
1	Belgium	471.1911	728.5526	257.3615	54.619347
2	Czechia	335.4563	551.0919	215.6356	64.281279
3	Denmark	664.0875	813.5892	149.5017	22.512350
4	Estonia	453.1148	383.1758	-69.9390	-15.435161

- **Create Metrics:**

1.2 Total waste generated	1.2 Recycling	% Recycling rate (%)
4731.7	1179.7	24.93%

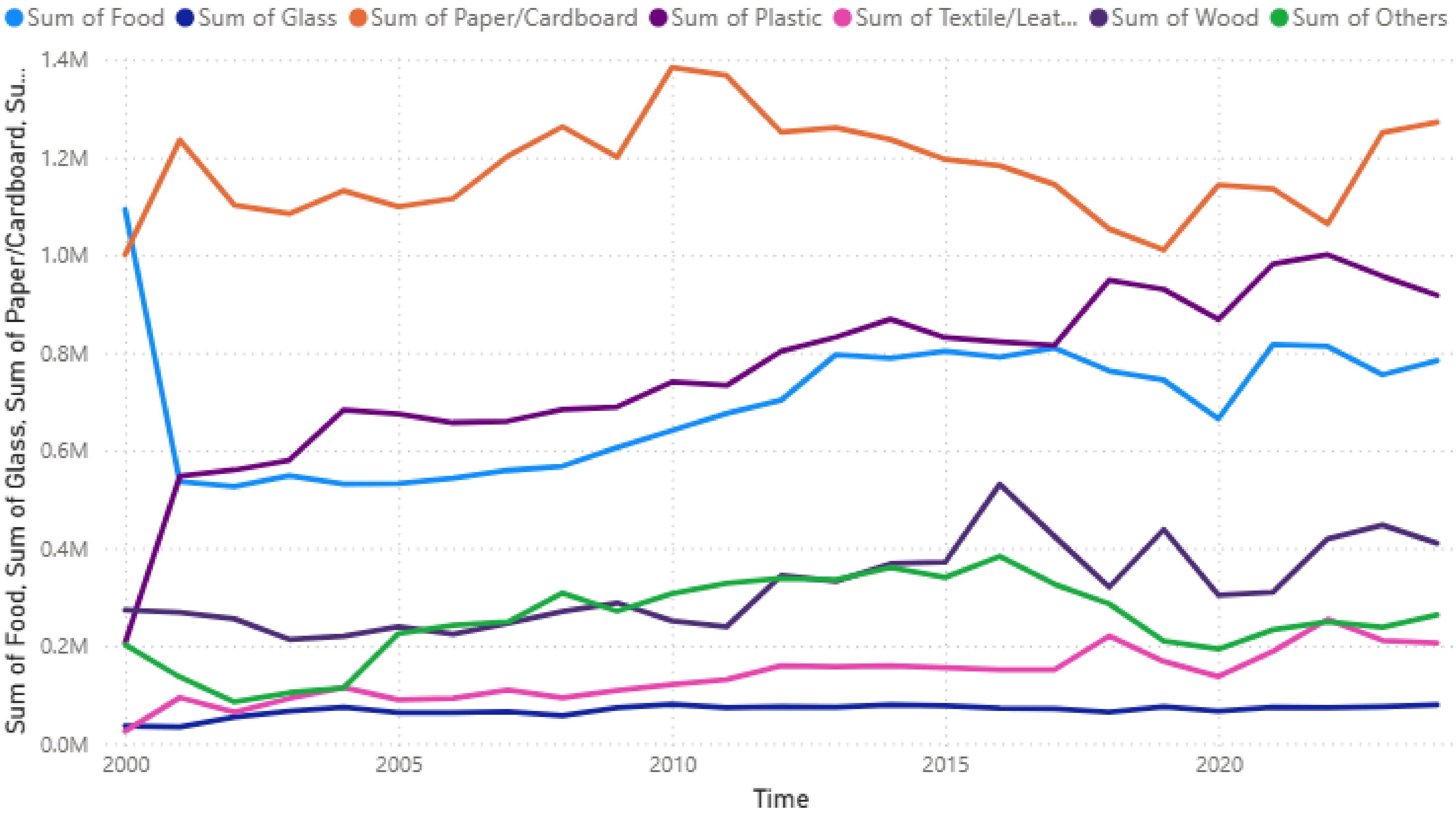
Using PowerBI edit query to calcuate percentages for further analysis

INDIVIDUAL DATA DASHBOARDS & INSIGHTS

Singapore Domestic Waste Analysis (2000-2024)

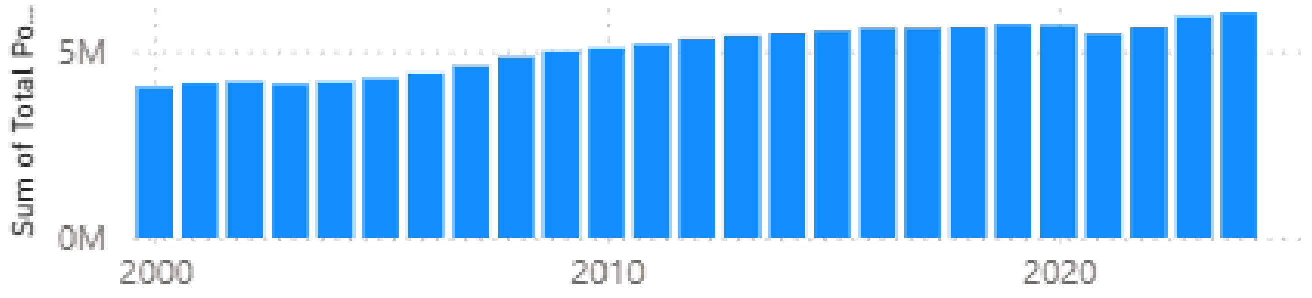
EMMANUEL

Sum of Food, Sum of Glass, Sum of Paper/Cardboard, Sum of Plastic, Sum of Textile/Leather, Sum of Wood and Sum of Others by Time



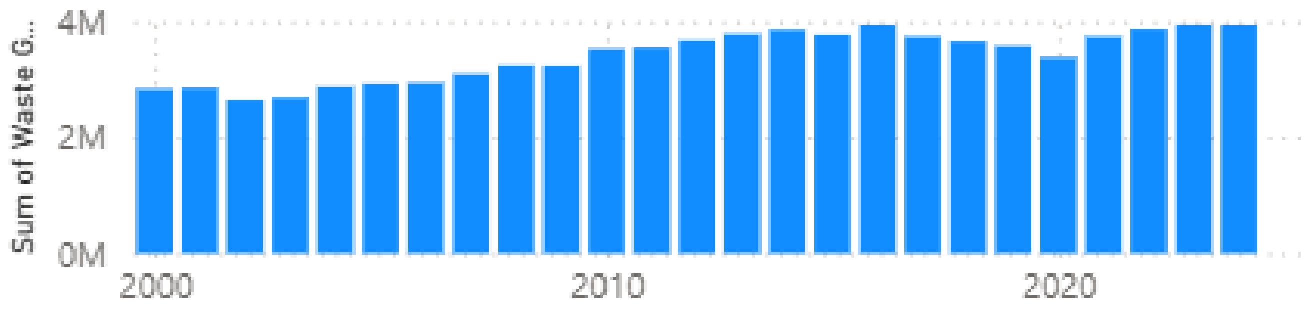
Waste Type Generated in Ascending Order: Glass, Textile/Leather, Others, Wood, Food, Plastic, Paper/Cardboard

Sum of Total Population by Time



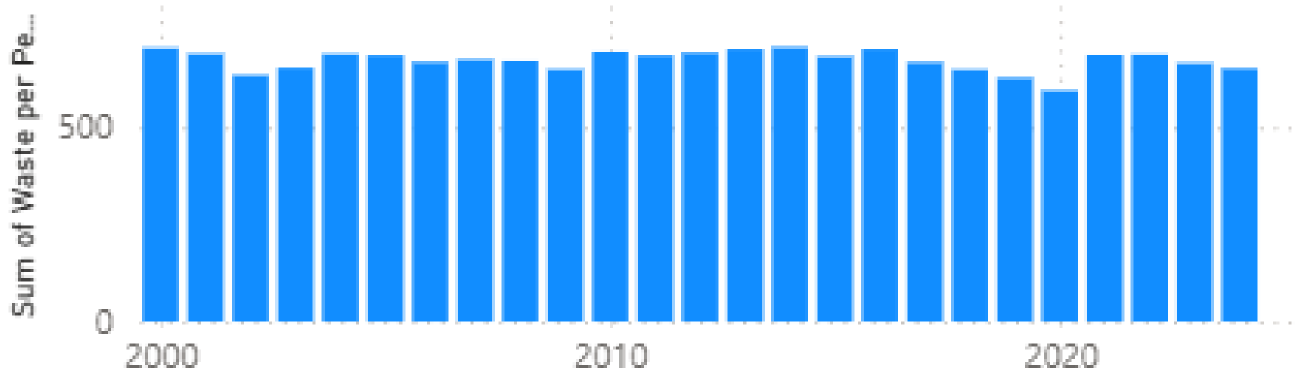
2000: 4,027,887 / 2024: 6,036,860

Sum of Waste Generated (ton) by Time



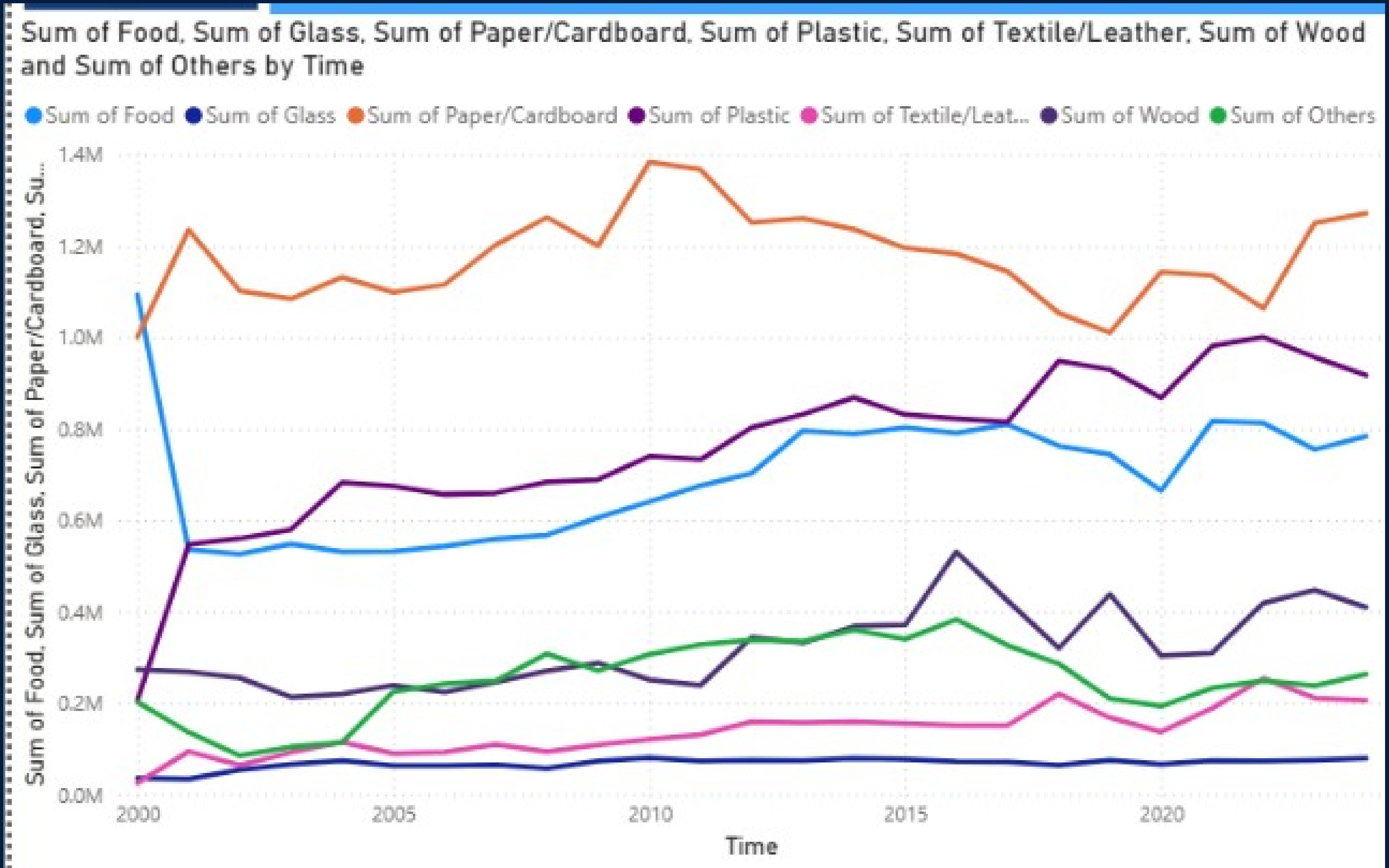
2000: 2,826,000 ton / 2024: 3,932,000 ton

Sum of Waste per Person (kg) by Time



2000: 704.09kg / 2024: 651.33kg

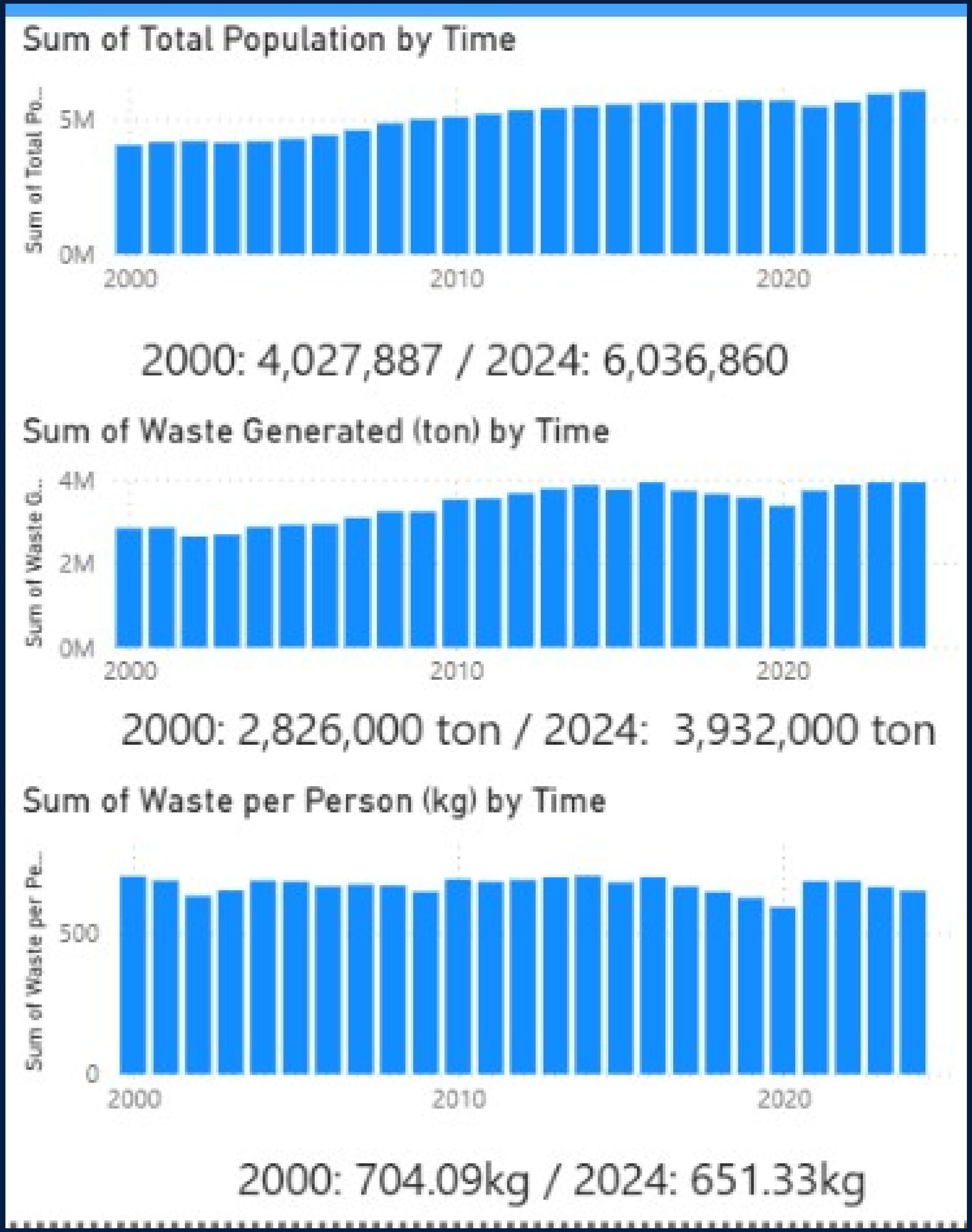
EMMANUEL'S INSIGHT



Waste Type Generated in Ascending Order: Glass, Textile/Leather, Others, Wood, Food, Plastic, Paper/Cardboard

Paper and Cardboard generated the most amount of waste averaging 1.2 Million Tonnes. In contrast Glass generated the least amount of waste, only a few thousand tonnes.

EMMANUEL'S INSIGHT



As the population of Singapore increases from around 4 Million to around 6 Million, The waste generated also increases from around 3 Million ton to around 4 Million ton. Surprisingly, the waste per person decrease over the years from around 700kg to around 650kg.

SINGAPORE RECYCLING ANALYTICS

Worse Recycled Type(%)
2024

4.70

Total Recycled
2024

849000

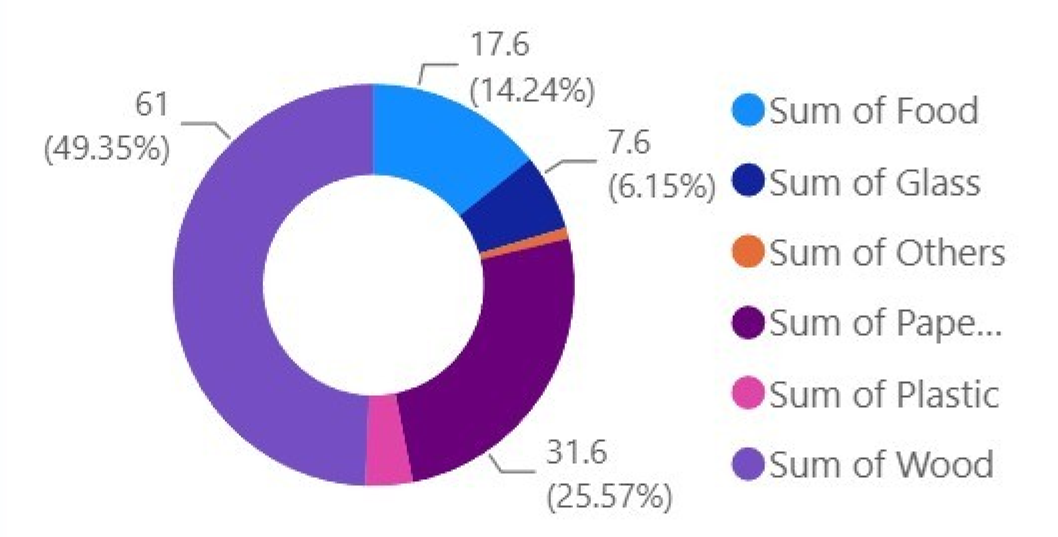
Total Disposed
2024

3084000

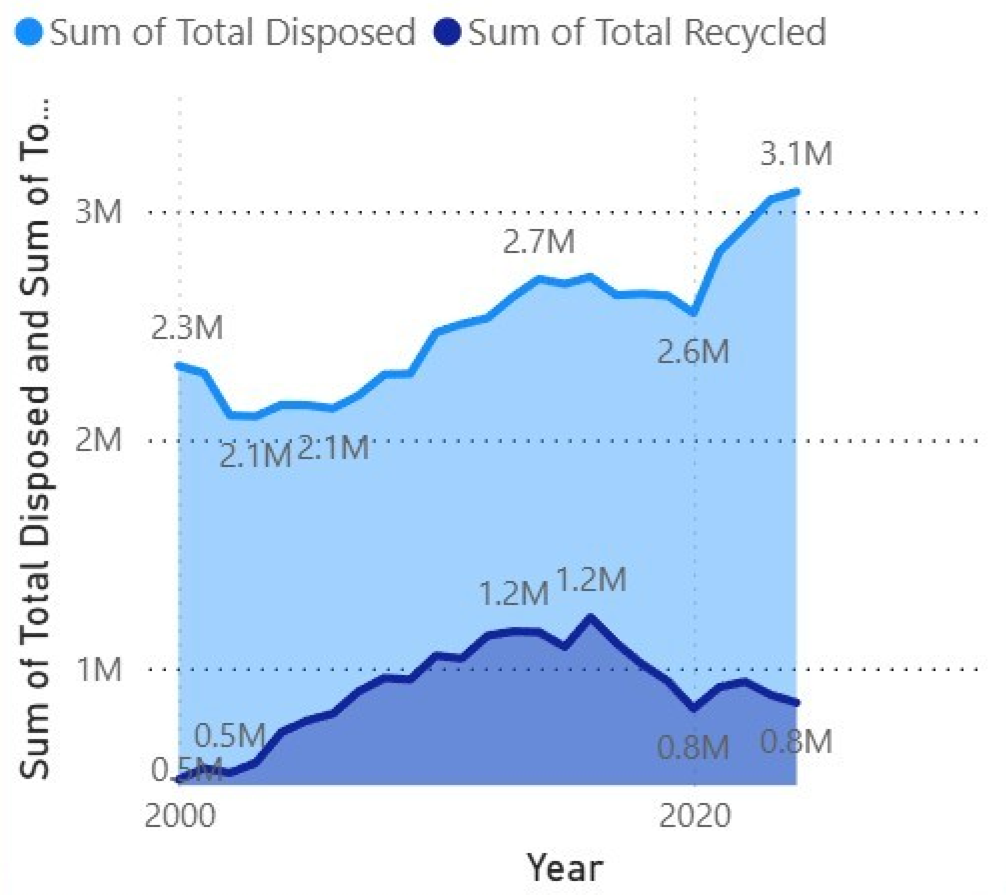
Recycling Rate(%)
2024

21.6

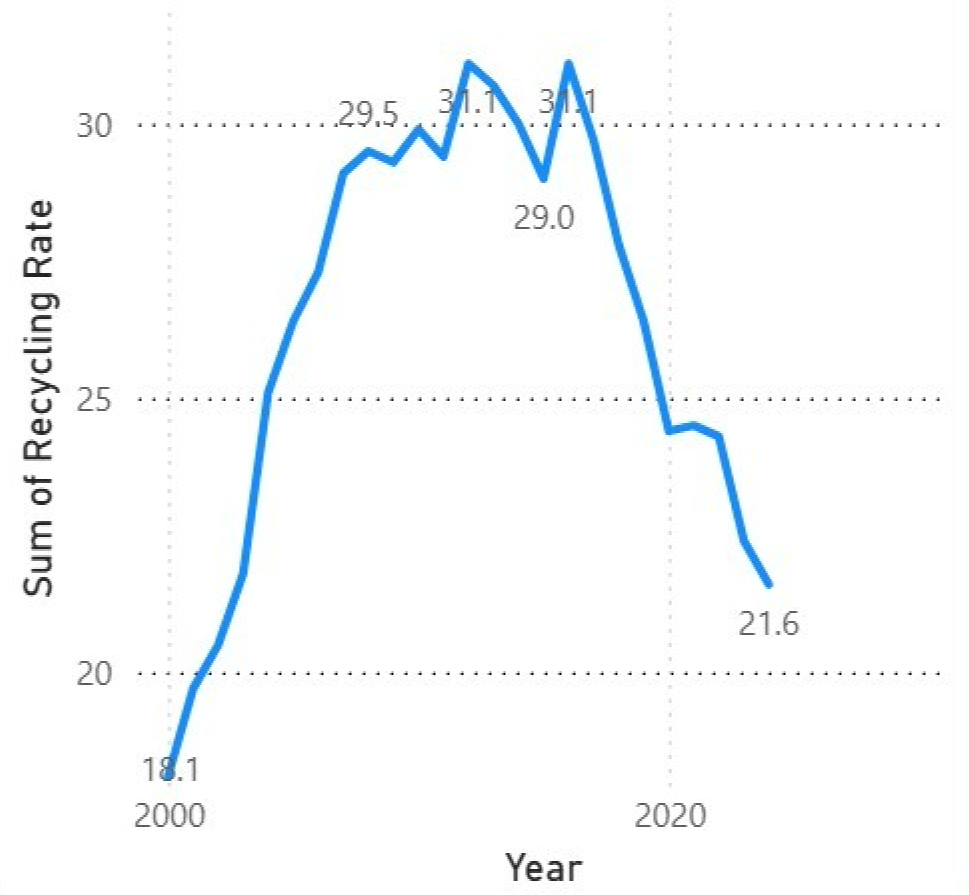
Recycling Rate By Type
2024



Disposed VS Recycled
(2000-2024)

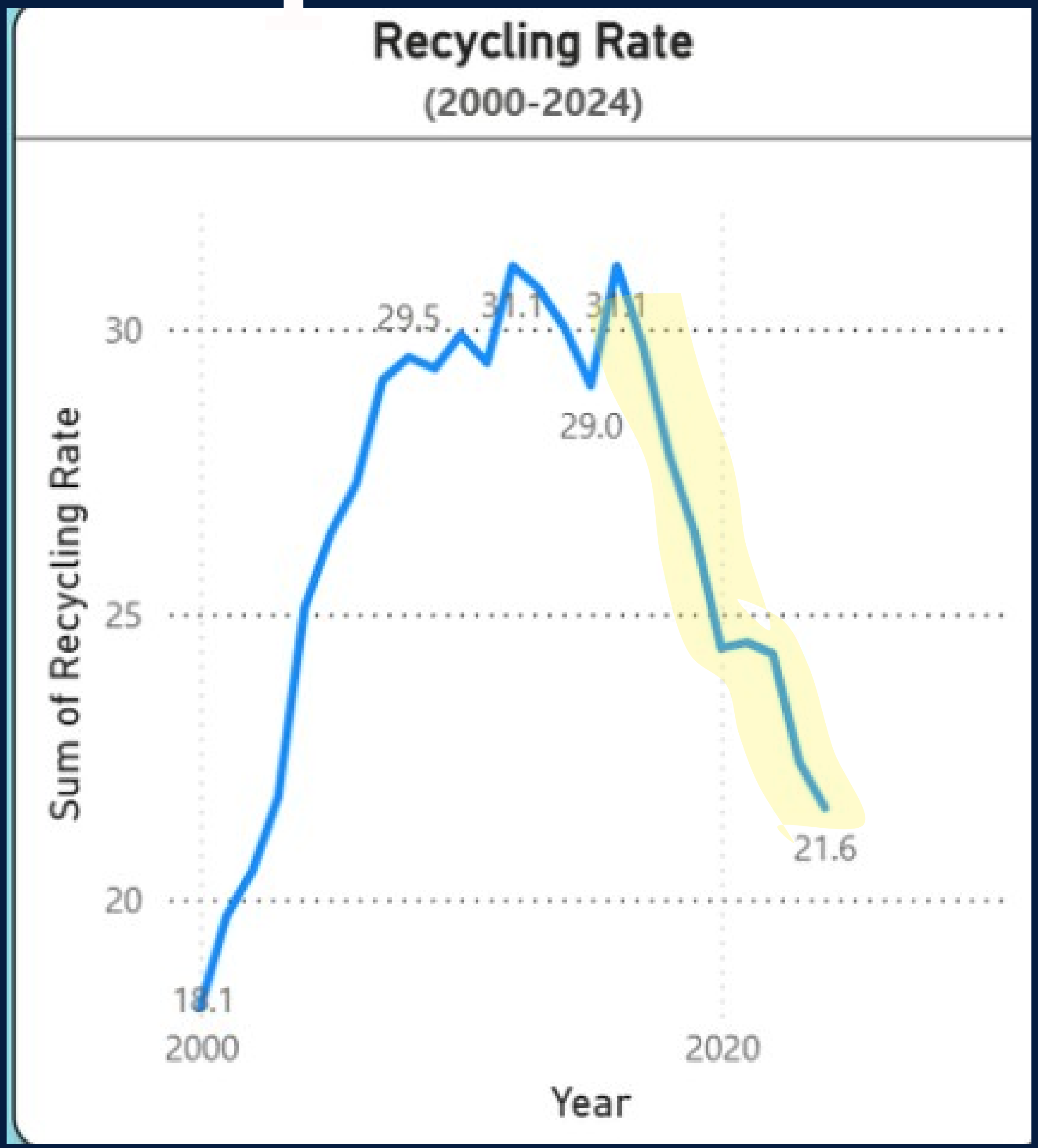


Recycling Rate
(2000-2024)



[PIERRE] — CAUSE AND EFFECT

INSIGHT 1



CAUSE

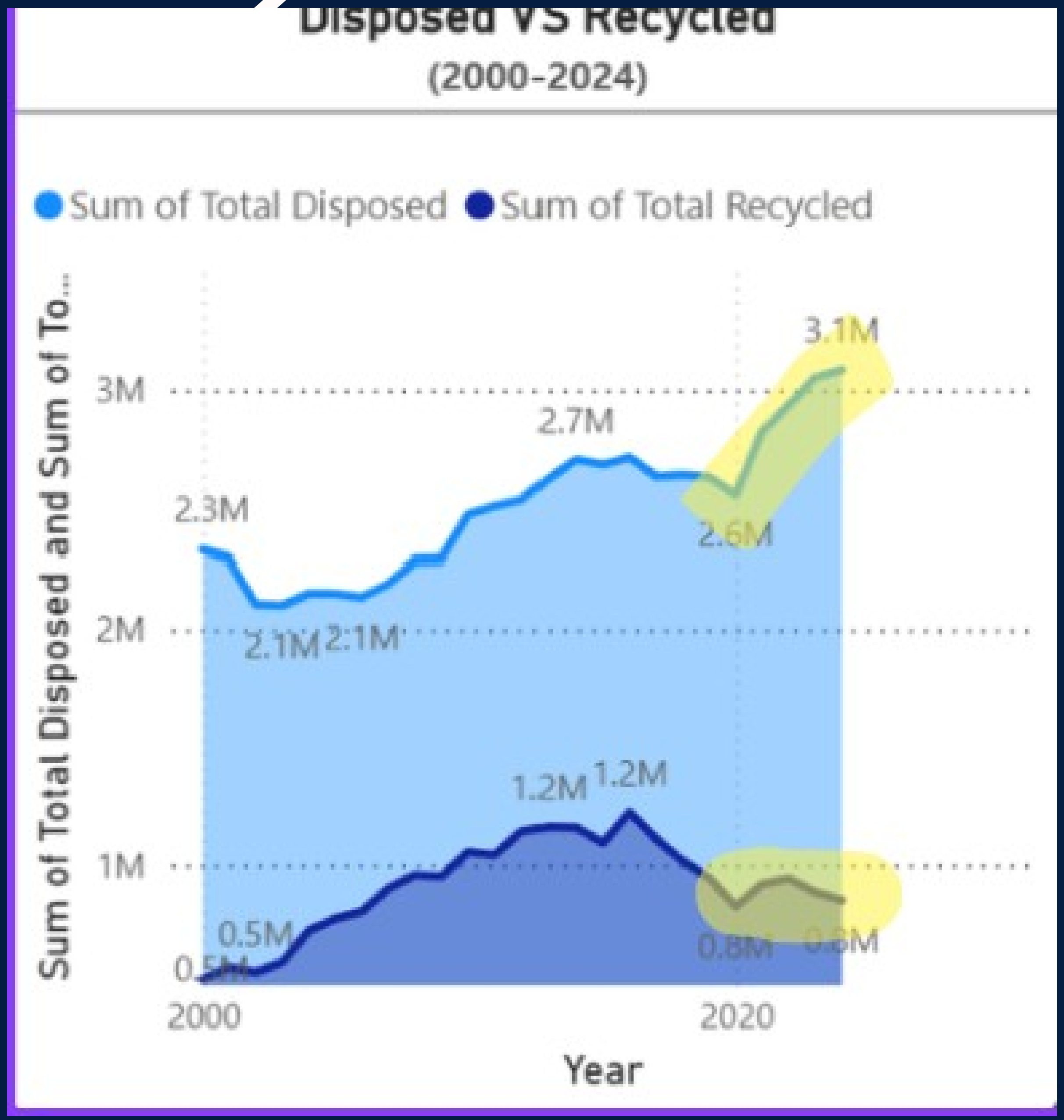
OVERALL REC POST-COVID FACTORS INCLUDING HIGHER FREIGHT COSTS AND WEAKER DEMAND FOR RECYCLED MATERIALS

EFFECT:

EFFECT: THE OVERALL RECYCLING RATE FELL FROM A PEAK OF ~31.1% (AROUND 2018-2019) TO JUST 21.6% IN 2024 — A DROP OF ROUGHLY 10 PERCENTAGE POINTS IN 5-6 YEARS

[PIERRE] — CAUSE AND EFFECT

INSIGHT 2



CAUSE:

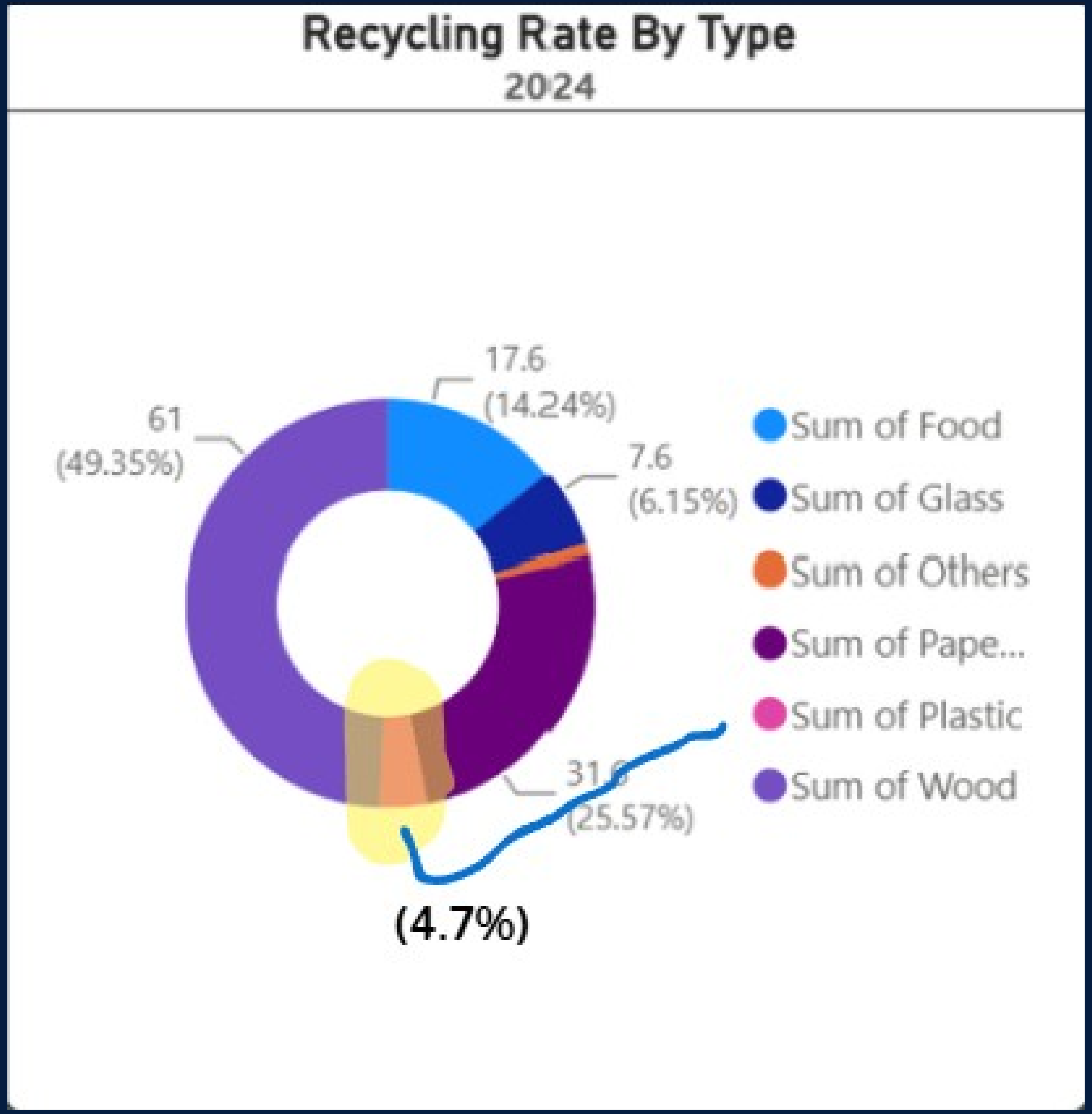
DRIVEN MAINLY BY CONSTRUCTION & DEMOLITION WASTE

EFFECT:

EFFECT: TOTAL DISPOSED VOLUME ROSE SHARPLY FROM ~2.6M TO 3.1M TONNES IN RECENT YEARS, WHILE TOTAL RECYCLED STAYED ROUGHLY FLAT AT ~0.8M TONNES

[PIERRE] — CAUSE AND EFFECT

INSIGHT 3



CAUSE:

ONLY 18% OF FOOD WASTE WAS RECYCLED IN 2024

EFFECT:

FOOD IS ONE OF THE LARGEST WASTE CATEGORIES GENERATED IN SINGAPORE BUT HAS THE LOWEST RECYCLING RATE ON THE DASHBOARD, AT JUST 4.70%

(4.7 %)

KWAN TENG

Singapore's Waste Generate per Capita and Recycling Rate Forecast

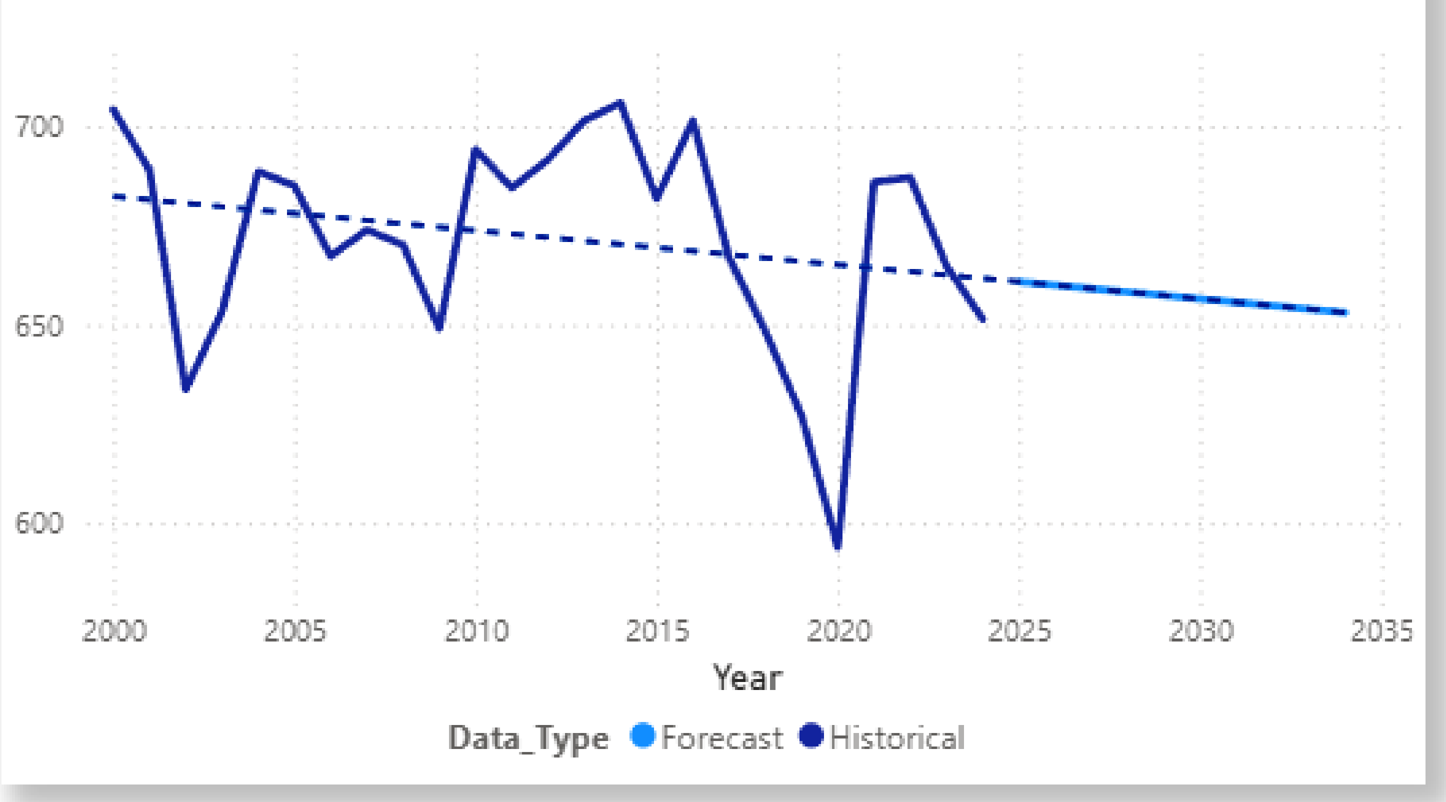
Latest Forecast Waste Display

653.05 kg/person

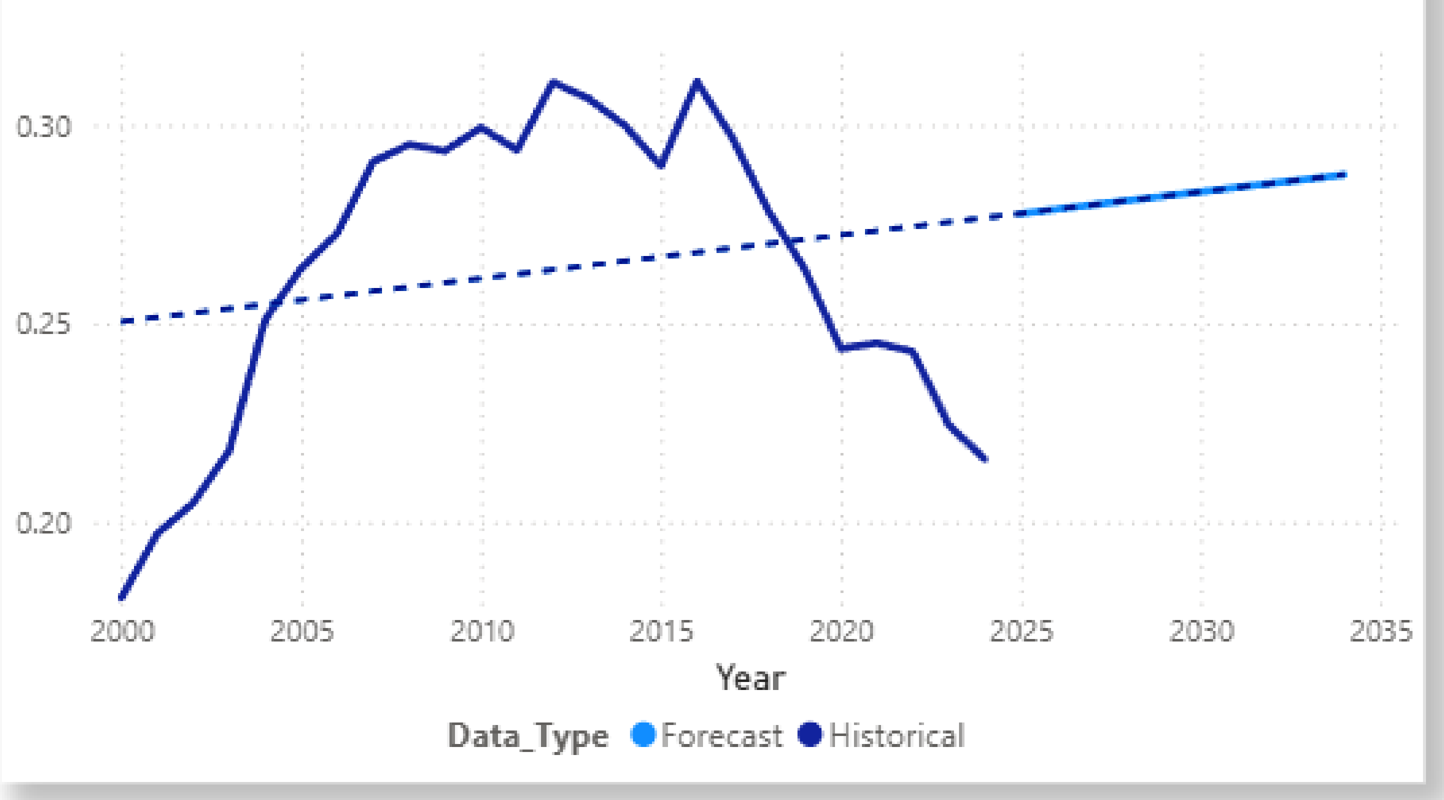
Latest Forecast Recycling Rate

28.75%

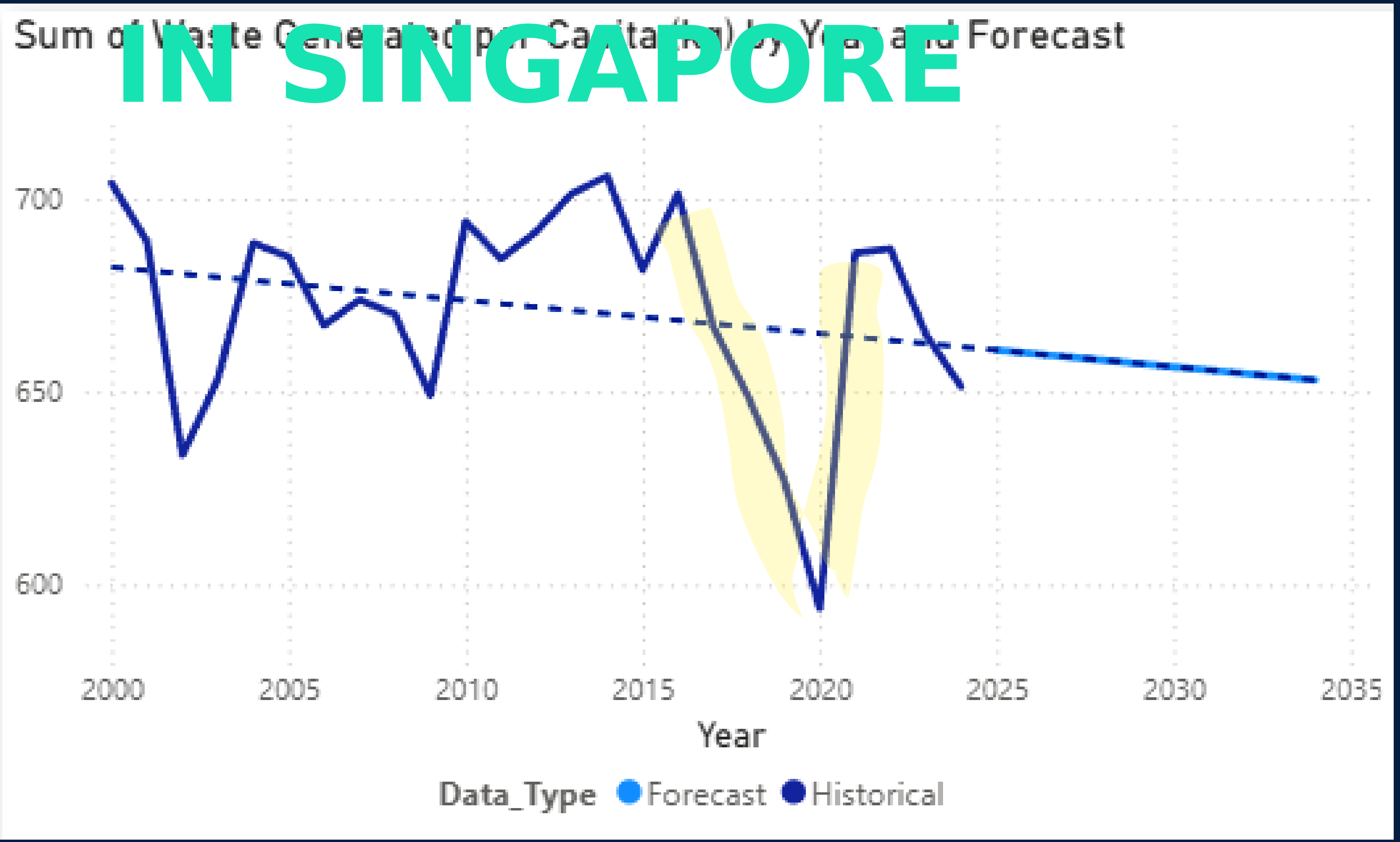
Sum of Waste Generated per Capita (kg) by Year and Forecast



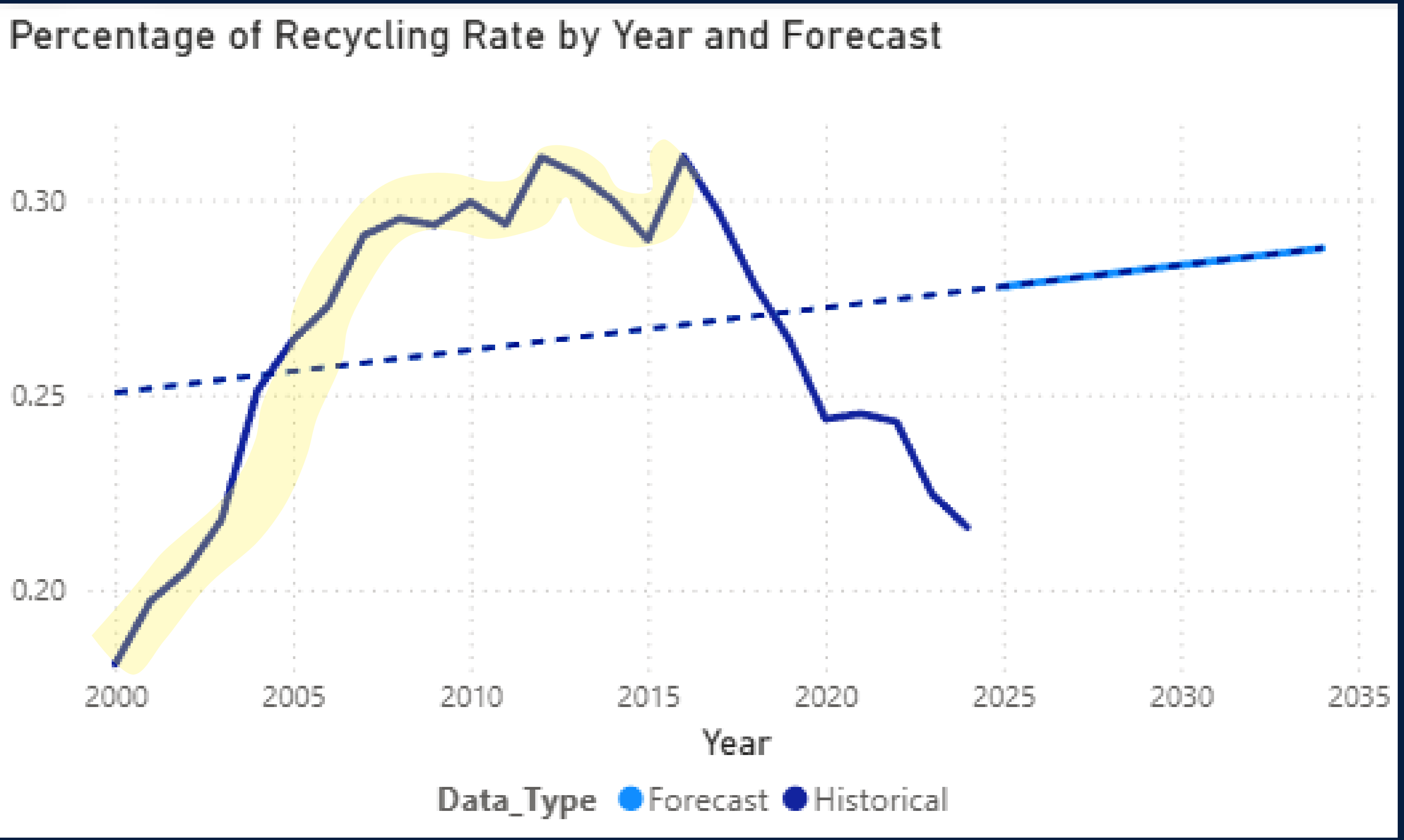
Percentage of Recycling Rate by Year and Forecast



KWAN TENG — FORECAST OF WASTE GENERATED PER CAPITA AND REYCLING RATE IN SINGAPORE



Due to the huge decrease in 2020 from the Covid-19 pandemic. It will shows a decreasing trend.



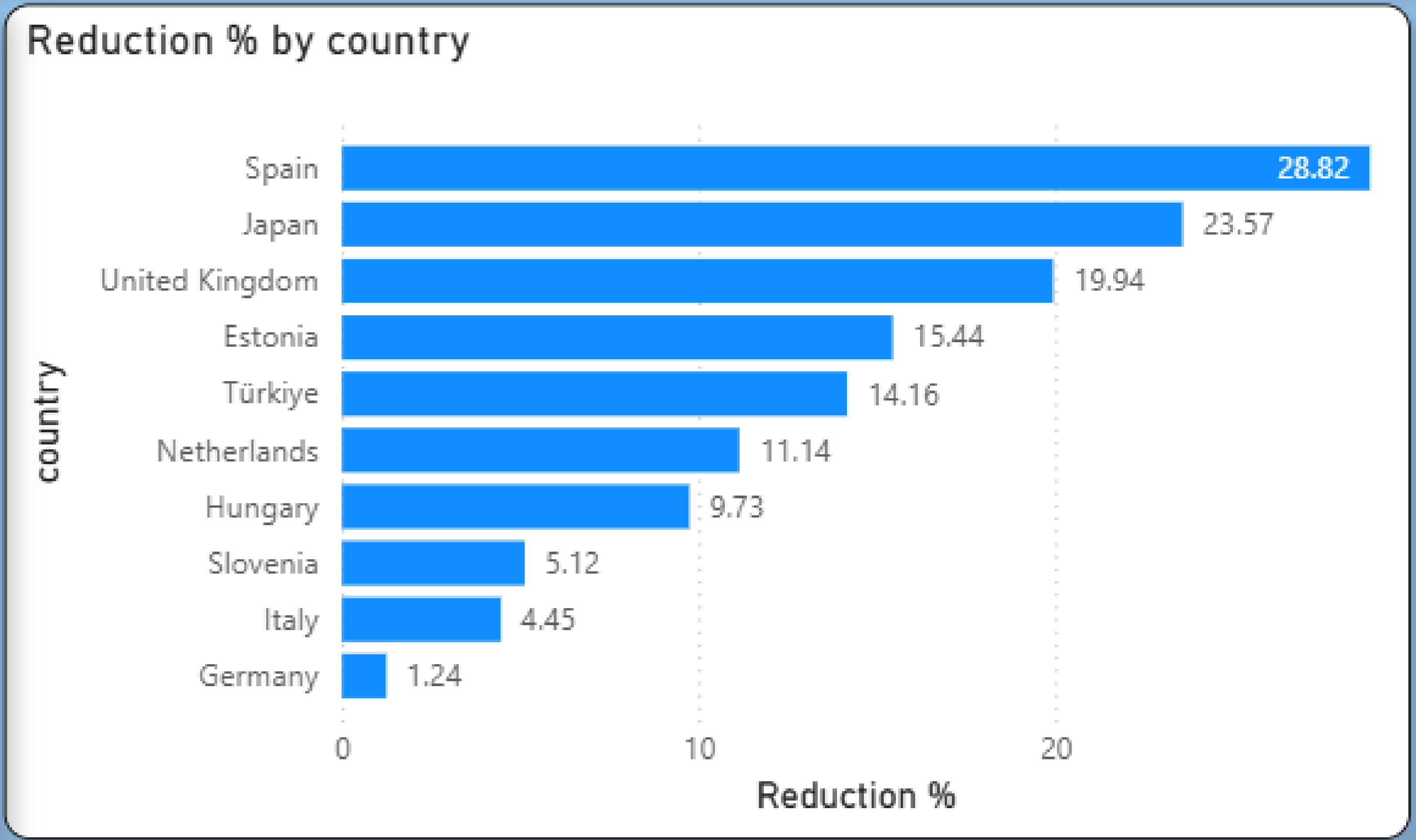
As there is an increasing trend until 2016, the trendline shows a increasing trend.

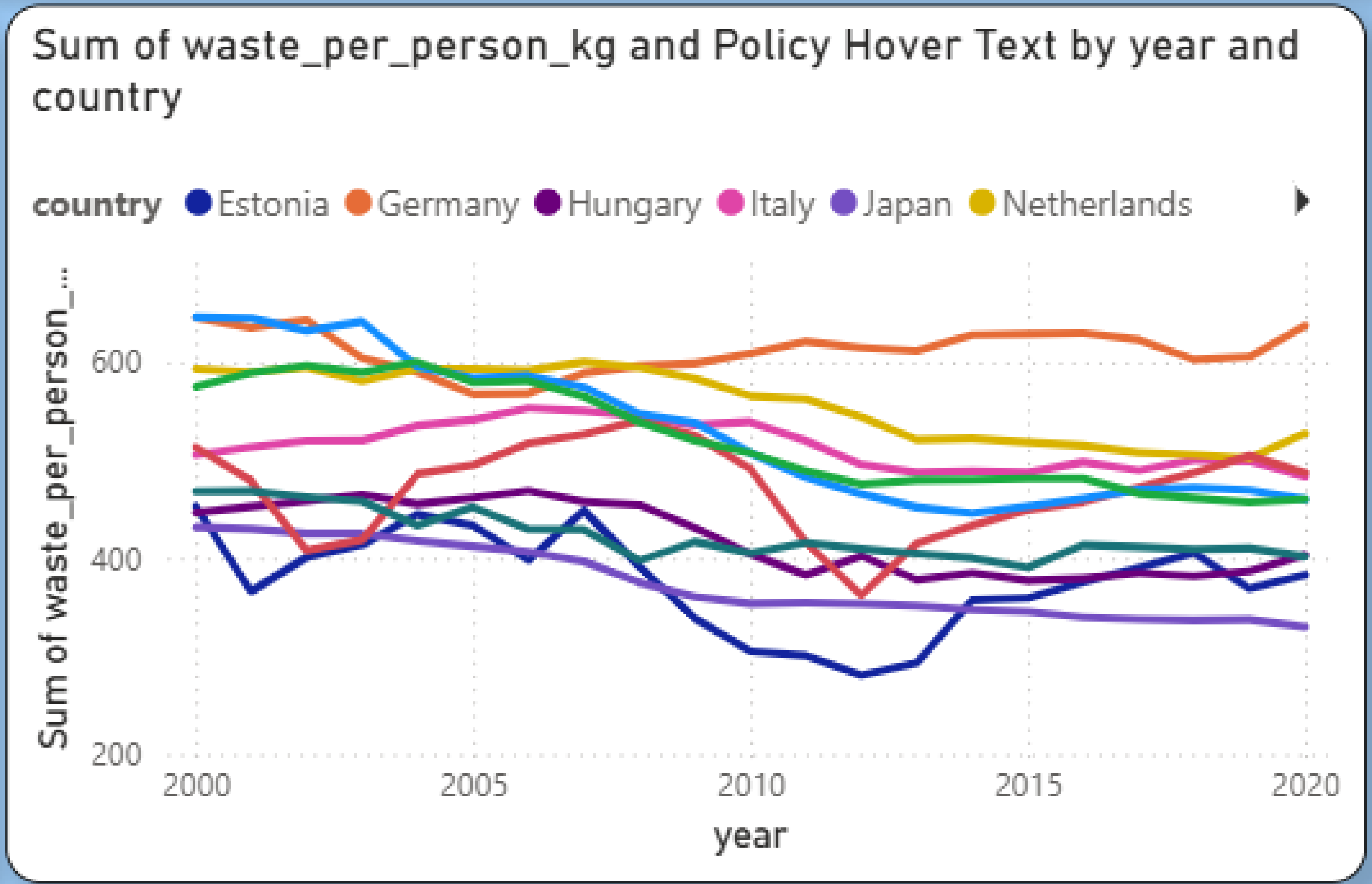
ZI QIAN — DASHBOARD 4:
OECD WASTE BENCHMARKING & POLICY SCENARIO

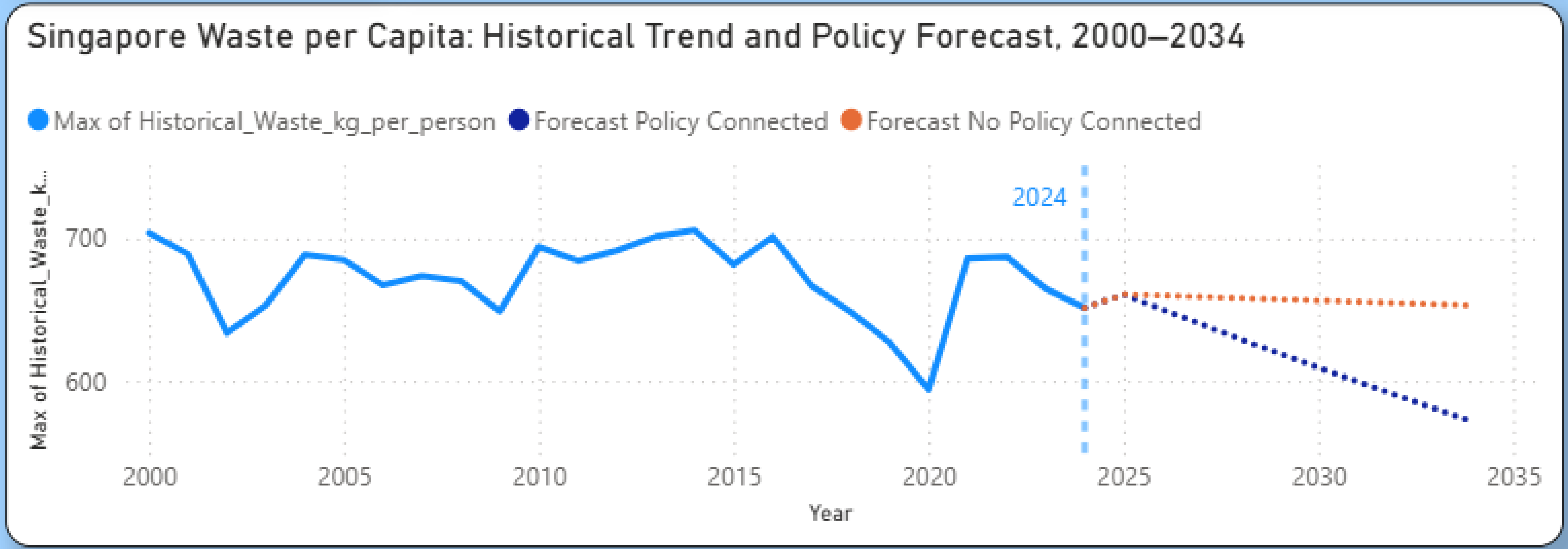
Singapore Waste Reduction: OECD Benchmarking and Policy Scenario

Top 3 country

- [ ] Japan
- [ ] Spain
- [ ] United Kingdom







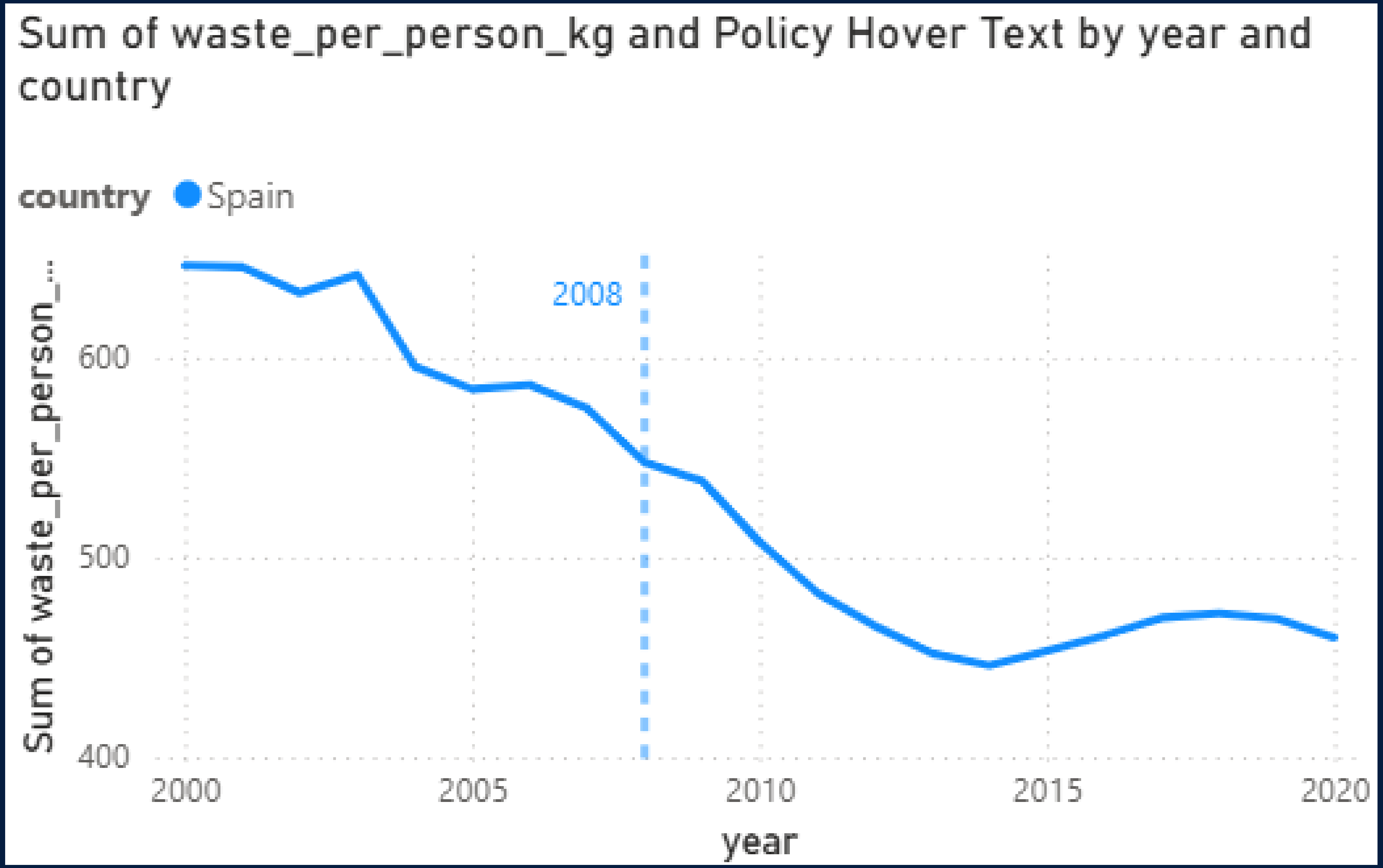
571.03
Forecast With Policy 2034

12.56%
Policy Reduction 2034

653.05
Forecast Without Policy 2034

82.02
Waste_Prevented 2034

Why Did Spain’s Waste per Person Drop So Much?



POLICY — 2008

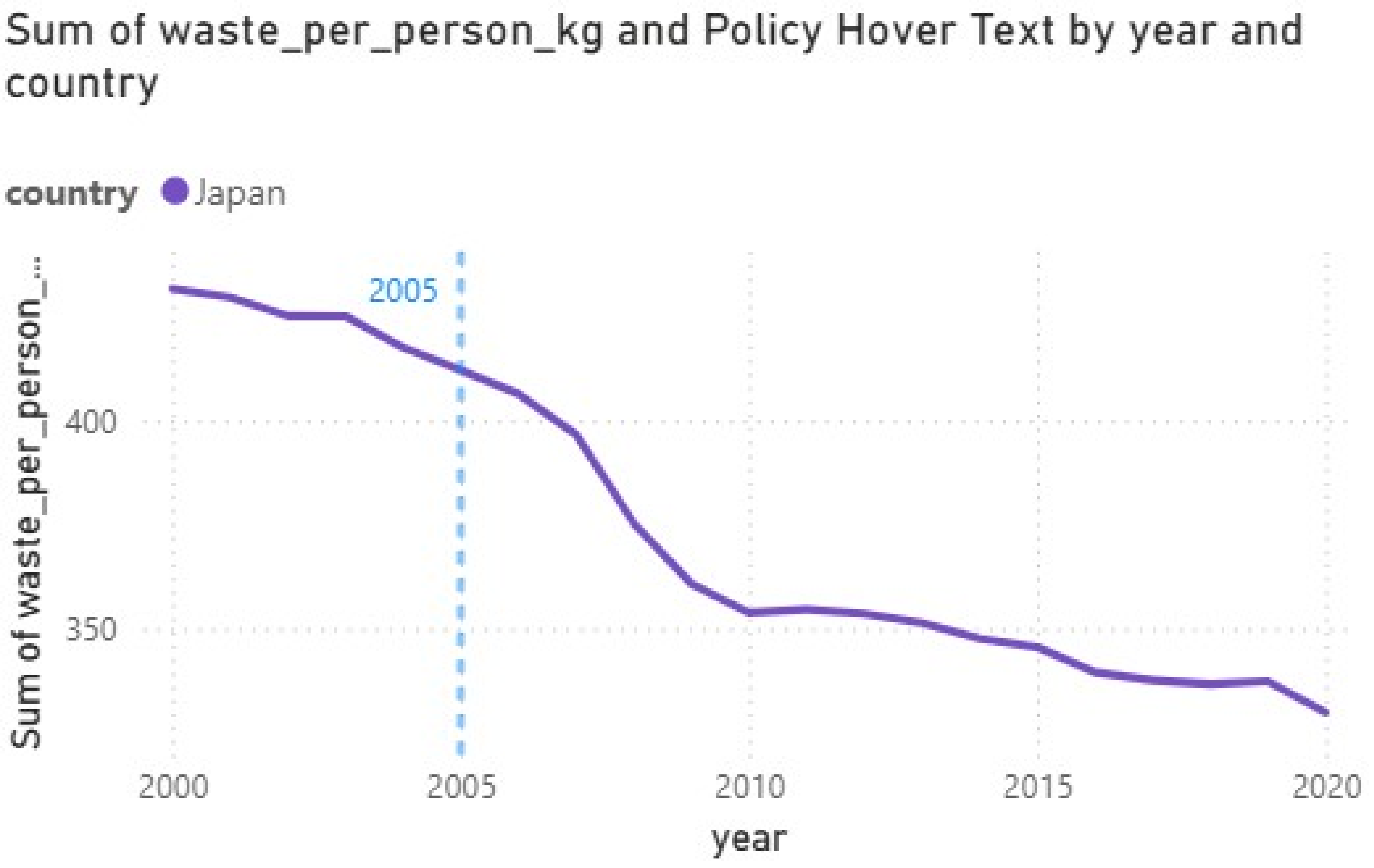
Single-Use Shopping Bag Reduction Measures

What it did:

Targeted 50% fewer single-use shopping bags.

Worked with retailers to reduce bag use and promote reusable bags.

Why Did Japan’s Waste per Person Drop So Much?



POLICY — 2005

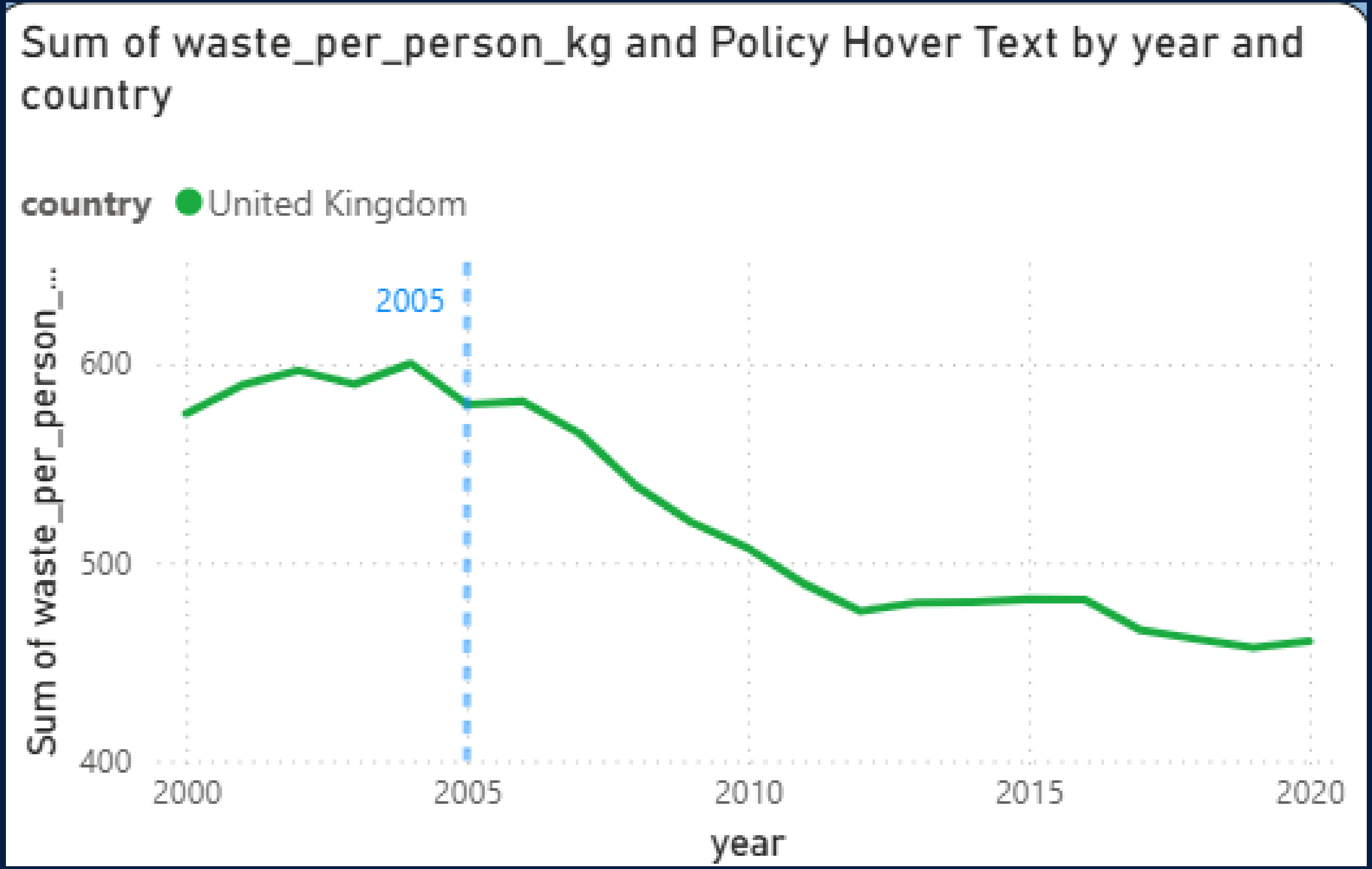
Waste Collection Charging Policy

What it did:

Municipalities were encouraged to charge people for waste collection.

The idea was simple: if throwing away more waste costs more money, people have a reason to reduce waste.

Why Did United kingdom's Waste per Person Drop So Much?



POLICY — 2005

Courtauld Commitment — Phase 1

What it did:

Major supermarkets, food companies and manufacturers agreed to reduce household food waste and unnecessary packaging waste.

OUR SOLUTION FOR WASTE REDUCTION:

SINGAPORE RECOMMENDATION
RETURN & SAVE TAKEAWAY SCHEME

A reusable takeaway-container deposit system

PAY DEPOSIT
Pay a $2 refundable deposit

USE
Enjoy your meal using the reusable container.

Return
Return the container at the return machine

GET $2 BACK
Receive the full deposit back after returning the container.

Why It Can Work in Singapore

👥
Consumers
Get their money back by returning the container.

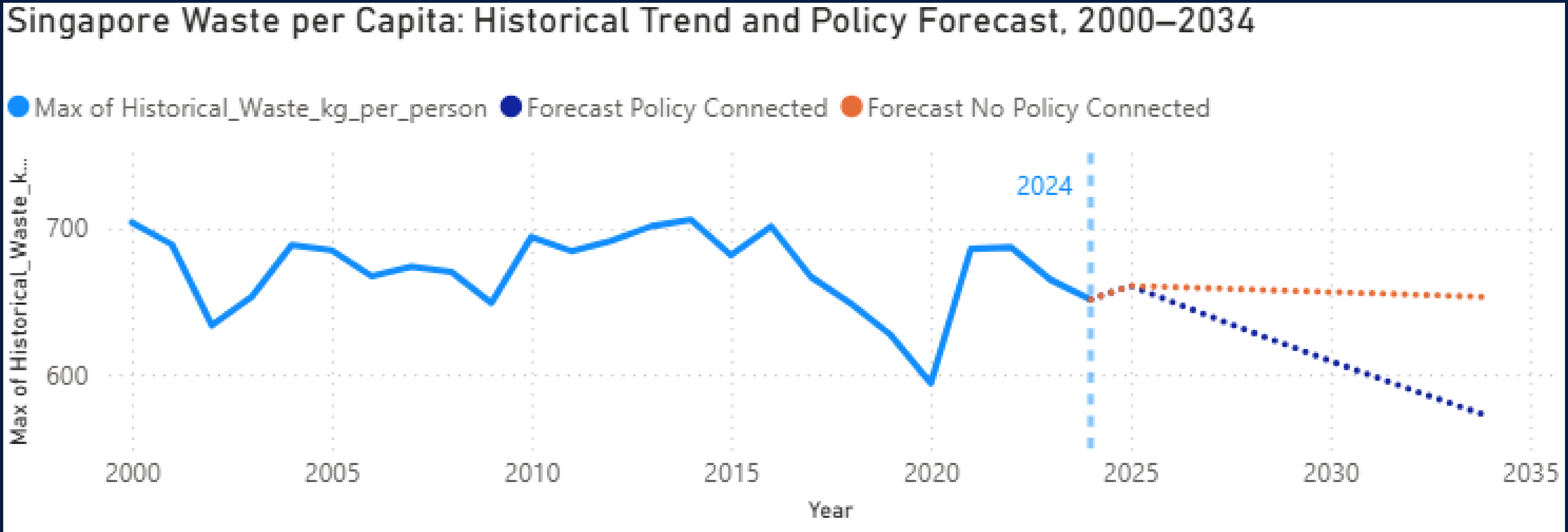
🏪
Businesses
Use the same standard containers and shared return system.

🏛️
Government
Builds on Singapore's existing return-system experience.

❤ More job:

people is need to wash and clean the container

FUTURE FORCAST



571.03
Forecast With Policy 2034

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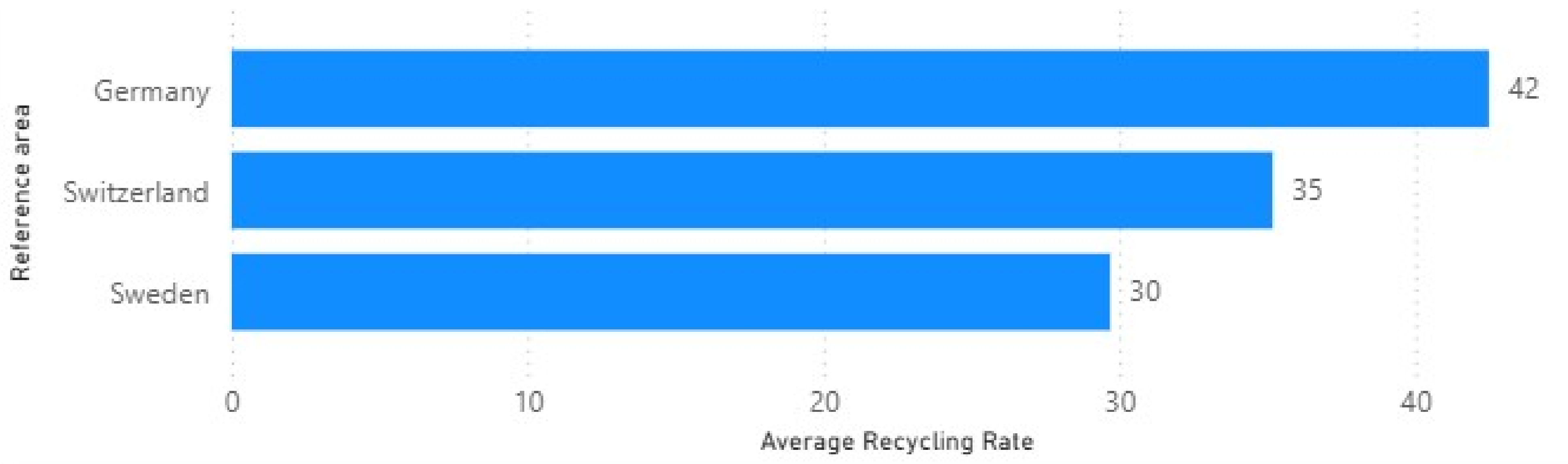
Waste_Prevented 2034

Hong Jean - Overseas Countries recycling statistic

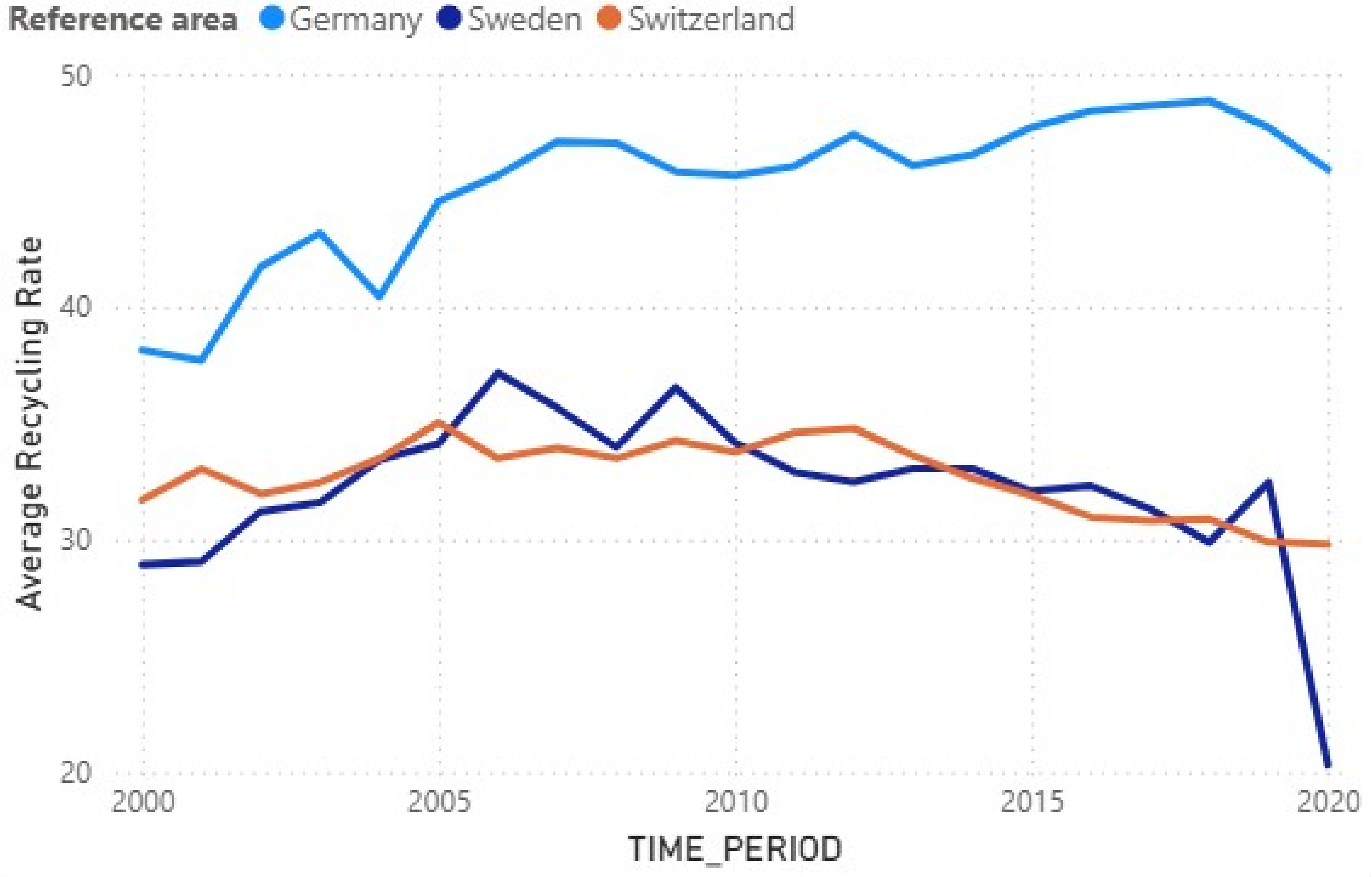
Overseas Country's Recycling Statistics

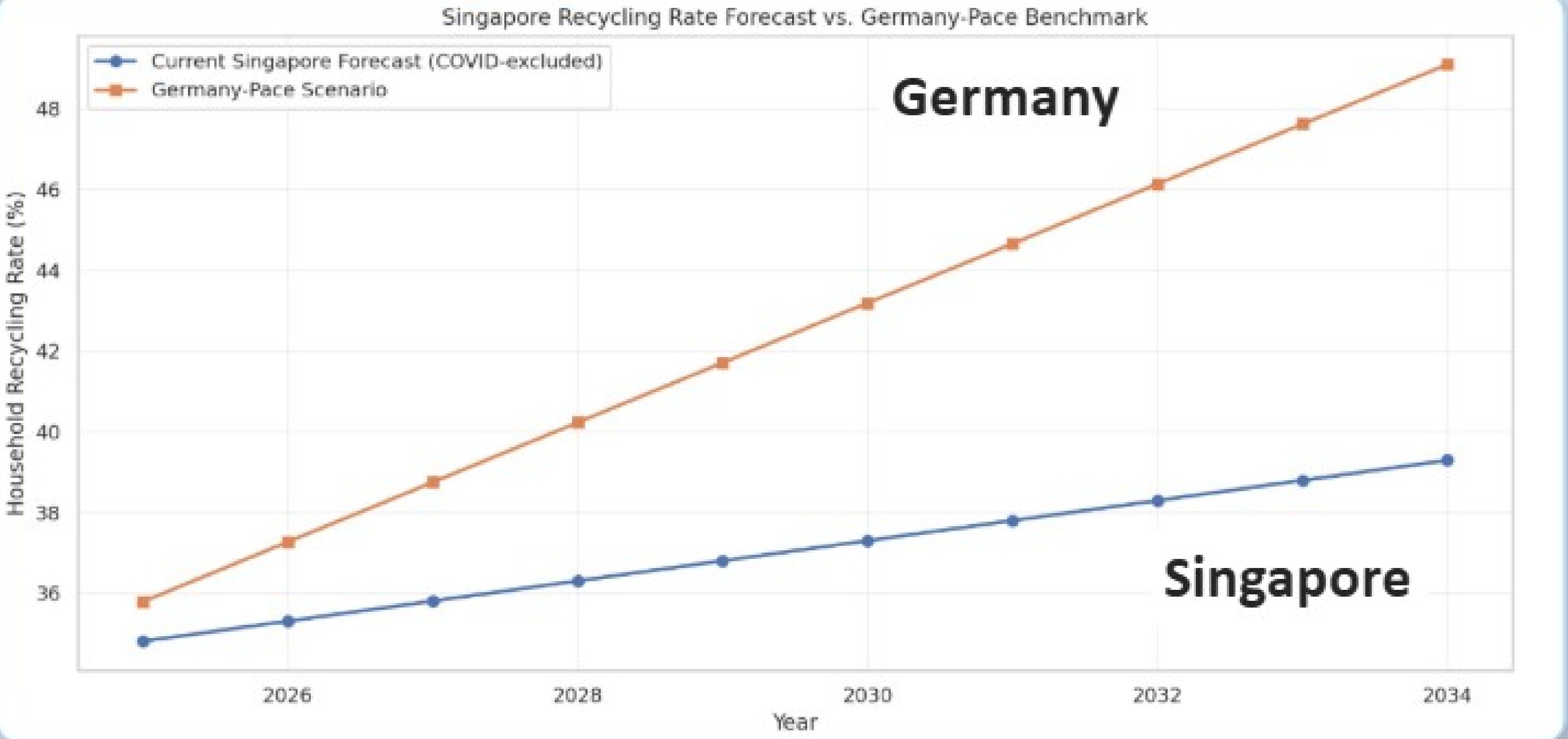
Germany Overall Recycling Rate (%)	Switzerland Overall Recycling Rate (%)	Sweden Overall Recycling Rate (%)
42.49%	35.18%	29.69%

Top 3 Country's Overall Recycling Rate (%) (2000-2020)

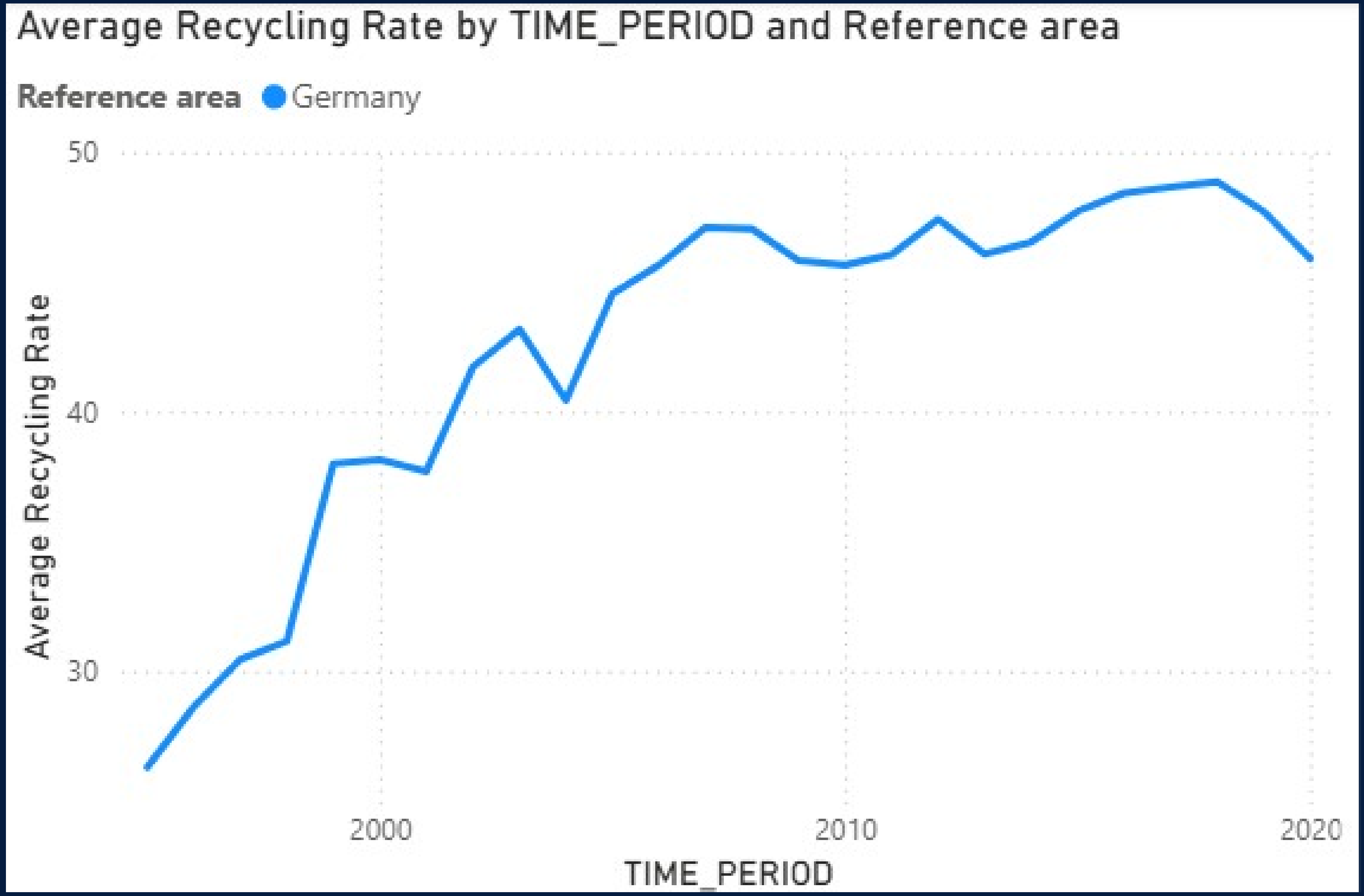


Top 3 Country's Recycling Rate Trend (%) (2000-2020)





Packaging Ordinance - 1991



- **Producer Responsibility:**

Makes companies liable for their packaging waste.

(Singapore already had this producer responsibility policy)🚫

- **Separated Collection:** ⭐

Kept separate recyclable materials from general household waste.

Solution Comparison Analysis

Singapor
e

Recycle blue bin under HDB:

“Does people willing to go all the way down to dispose their recycling material? “

Germany

Yellow bag

packaging waste is collected separately from normal household waste,

Whch can be left outside their house for people to collect

RECOMMENDED RECYCLING SOLUTION

Adopt Germany's Waste seperation Method

Dedicated Yellow Bag:
Provided to every household for clean, recyclable materials.

Convenient Doorstep Collection:
Simply leave bags outside the front door for pickup.

Contamination Control:
Visual checks ensure materials meet recycling standards before collection.

Reward system :
if certain quota being met for each household ,

CONCLUSION

LESS WASTE
SENT TO PULAU SEMAKAU

THANK YOU

“Lets make singapore green again”